CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (CAD/CAM) Sem - I University Examination, 2009-10

CE – 725 Artificial Intelligence

Date: 29.12.09, Tuesday Time: 01:00 p.m To 04:00 p.m Maximum Marks: 70

Instructions: (1) All questions are compulsory.
(2) Indicate clearly, the option you attempt along with its respective question number.
(3) Figures to right indicate marks
(4) Rough work is done in the last page of main supplementary, don't write anything on the question paper.

SECTION-I

Q-1 Answer the following questions.

1. What is artificial intelligence? Give comparisons between human and computer intelligence? 3
2. Draw a state-space graph for the water jug problem. 3
3. Differentiate between non monotonic reasoning and non monotonic logic. 2
4. Give the difference between Best First Search and Hill Climbing technique. 3

Q-2

[A] **State True or False** 4
1. Hill climbing is similar to generate and test in which feedbacks are used.
2. Backtracking is not a solution to local minima problem.
3. If a heuristic value overestimates then A* guarantees an optional solution.
4. Resolution produces proof by contradiction.

[B] Write a PROLOG program to find the maximum and minimum number from a set of n numbers. 4

[C] Construct partitioned Semantic Net representation for the knowledge 4
Every batter hits a ball.

OR

[A] Explain: Multilayer Feed forward Networks. 4

[B] What is an Expert System? Explain the role of knowledge in design of Expert System. 4

[C] Can a single neuron learn a task? How does the perceptron learn its classification tasks? 4

Q-3

[A] Assume the following facts: 4
(i) Steve only likes easy courses.
(ii) Science courses are hard.
(iii) All the courses in the basket weaving department are easy.
(iv) BK301 is a basket weaving course.
Convert the above facts into clausal form and use resolution to answer the question, "What course would Steve like?"

[B] Show how means-ends analysis could be used to solve the problem of getting from one place to another. 4

[C] Discuss any two real life applications of AI. 4

OR

[A] "Best first Search algorithm is not adequate for searching AND-OR graphs" – Discuss with an example. 4

[B] How does Iterative Deepening improve the performance of A* search algorithm. 4

[C] Write a PROLOG program to reverse a list. 4

SECTION-II

Q-4

1. Why natural language processing is required? What are the issues in natural language understanding? 3
2. Derive the parse tree for the sentence using the appropriate rules: **John loves marry.** 4
3. Differentiate between Frames and Semantics net. Give examples. 4

Q-5

[A] Write Unification algorithms and Trace the operation of the unification algorithm on each of the following pairs of literals: 4

(i) f(Marcus) and f(Caesar)
(ii) f(x) and f(g(y))

[B] Enlist any two application areas, where Artificial Neural Network will be preferably applicable. 4

[C] Justify the role of Genetic Algorithms and Fuzzy logic as soft computing components. What are its uses? 4

OR

[A] Give three situations where cut and fail might be useful. 4

[B] Construct semantic net representations for the following: 4
Mary gave the green flowered vase to her favorite cousin.

[C] A problem solving search can proceed either in forward or backward. Whether search should proceed forward or backward for the following problems. Justify the answer with the reason. 4

1. Medical Diagnosis
2. Natural language Understanding

Q-6

[A] Explain the Semantic Analysis and Discourse Integration Phases of NLP. 4

[B] Explain the working of Bayesian Network with an example. Also show the use of Bayes' Theorem in it. 4

OR

[B] Solve the following crypta arithmetic problem. 4

$$\begin{array}{r} TWO \\ +TWO \\ \hline FOUR \end{array}$$

[C] Explain how learning is achieved in neural network (how neural networks learn?). 4

OR

[C] Discuss about various components of an Expert System with a neat diagram. 4